

RAHUL KP

AI/ML Engineer | Computer Vision | Deep Learning | Generative AI | ML Systems

Kerala, India · Open to Remote / Relocation

[Portfolio](#) · [GitHub](#) · [LinkedIn](#) · [Hugging Face](#) · [ORCID](#) · [Google Scholar](#)

Email: rahulkpcurup@gmail.com

SUMMARY

AI/ML engineer with an MSc in Computer Science and a research-driven approach to building machine learning systems from mathematical foundations through production deployment. Experienced in implementing neural networks from scratch with NumPy, benchmarking PyTorch implementations, developing computer-vision and recommendation systems, and building production-oriented AI applications with FastAPI, PostgreSQL, Docker, and modern MLOps tooling. Research interests include deep learning, computer vision, recommender systems, generative AI, LLMs, and efficient model deployment.

EDUCATION

MSc Computer Science — University of Calicut

2024 – 2026 | ~77%

- Advanced coursework and independent work in artificial intelligence, machine learning, deep learning, algorithms, and computer science.
- MSc research focused on Neural Collaborative Filtering and reproducible ML system development.

BSc Computer Science

2020 – 2023

RESEARCH & PUBLICATIONS

Neural Collaborative Filtering for Movie Recommendation

Research / Preprint · 2026

Preprint DOI: 10.20944/preprints202605.0449.v1

- Designed and implemented a Neural Collaborative Filtering (NCF) movie recommendation system using both a from-scratch NumPy implementation and PyTorch.
 - Implemented neural-network forward propagation and manual backpropagation to establish a mathematical baseline before framework benchmarking.
 - Evaluated GMF + MLP architectures on MovieLens 1M with implicit-feedback training, 1:4 negative sampling, leave-one-out evaluation, 99 negative candidates, Hit@10, and NDCG@10.
 - Achieved comparable recommendation quality between scratch and PyTorch implementations while obtaining approximately **12.8× faster epoch execution** with PyTorch.
 - Extended the research into a production architecture using FastAPI, PostgreSQL, Docker, Nginx, monitoring, and Kubernetes-oriented deployment.
 - Manuscript submitted to IEEE Transactions on Knowledge and Data Engineering (TKDE).
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SELECTED PROJECTS

NCF-Recommender-System — Research-to-Production ML System

Python · NumPy · PyTorch · FastAPI · PostgreSQL · Docker · Next.js

- Built an end-to-end neural recommendation system spanning mathematical formulation, from-scratch implementation, PyTorch benchmarking, evaluation, inference serving, and production architecture.
- Implemented the NCF training pipeline manually with NumPy before reproducing it in PyTorch/MPS.
- Integrated TMDb movie metadata/posters and developed a web-based recommendation experience.
- Production stack includes FastAPI model serving, PostgreSQL, Docker, Nginx, monitoring, and Kubernetes/Terraform deployment concepts.
- Model configuration: approximately **1.23M parameters**.

CNN Foundation — Image Classification

Python · NumPy · PyTorch · Computer Vision

- Implemented convolutional neural-network fundamentals and trained an image-classification model on MNIST.
- Achieved approximately **98.6% classification accuracy** while focusing on understanding convolution, feature extraction, optimization, and training dynamics.

PhishGuard — Machine Learning Security System

Python · Scikit-learn · FastAPI · Testing

- Developed a machine-learning URL phishing detector achieving approximately **95% accuracy** and **0.98 ROC-AUC**.
- Built a tested inference service with **222 automated tests** and approximately **98.7% code coverage**.
- Designed the project around reproducible preprocessing, model inference, API serving, and software-quality practices.

RankScript — Learning Management / Ranking Platform

JavaScript · React · Node.js · PostgreSQL

- Developed a learning platform with a weighted leaderboard system.
- Ranking model combines quizzes (40%), assignments (30%), lessons (15%), and learning streaks (15%).
- Focused on backend logic, data modeling, and transparent ranking computation.

TaskFlow — Full-Stack Productivity Platform

JavaScript · React · Node.js · Express · MongoDB

- Built a full-stack task-management application with authentication, task workflows, and persistent data management.
- Applied REST API and frontend/backend integration patterns.

TECHNICAL SKILLS

Machine Learning & AI

Python · NumPy · Scikit-learn · PyTorch · XGBoost · LightGBM · Random Forest · Neural Networks · Deep Learning · Computer Vision · Recommender Systems · NLP · Generative AI · LLMs · RAG

Deep Learning

ANNs · CNNs · RNNs · LSTMs · Transformers · Neural Collaborative Filtering · Representation Learning · Transfer Learning · Model Evaluation · Manual Backpropagation

Generative AI / LLM Engineering

Hugging Face Transformers · LangChain · RAG · LoRA · QLoRA · LLM Fine-Tuning · Prompt Engineering · Local LLM Inference

Software Engineering

Python · C · C++ · Java · JavaScript · TypeScript · SQL · REST APIs · FastAPI · Flask · Node.js · Express · React · Next.js

Data & Infrastructure

PostgreSQL · Redis · MongoDB · Docker · Kubernetes · Nginx · Git · GitHub Actions · Prometheus · Grafana · Terraform · Tailscale

AI Deployment

PyTorch MPS · Model Serving · ONNX concepts · Containerized Inference · Production ML Architecture · Local Apple Silicon AI/ML

CERTIFICATIONS

- **AI/ML for Geodata Analytics** — Indian Space Research Organisation (ISRO), IIRS
 - **IBM Generative AI Engineering Professional Certificate**
 - **Oracle Cloud Infrastructure 2025 Generative AI Professional**
 - **20+ additional technical certifications** spanning AI/ML, software engineering, cloud, and related technologies
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RESEARCH & ENGINEERING INTERESTS

- Computer Vision and Deep Learning
 - Computational Photography
 - Efficient AI / Edge AI
 - Neural Recommendation Systems
 - Generative AI and LLM Systems
 - Model Optimization and Deployment
 - ML Systems and MLOps
 - Research-to-Production AI
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ADDITIONAL

- Strong preference for first-principles learning: **mathematics** → **from-scratch implementation** → **framework benchmarking** → **production deployment**.
- Comfortable working across research, model development, backend engineering, and deployment infrastructure.
- Interested in AI systems where algorithmic research directly translates into user-facing products.